

Corridor-Preserving Path Smoothing and Turning-Maneuver Optimization for Differential-Drive Mobile Robots

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Abstract—Global planners in ROS 2/Nav2 commonly produce piecewise-linear paths with abrupt heading changes, resulting in high curvature peaks and reduced motion continuity for differential-drive mobile robots. This paper proposes PSTMO, a path-smoothing and turning-maneuver optimization method. In contrast to greedy line-of-sight shortcuts, which may remove anchors that encode the planner corridor and approach direction, standalone PSTMO does not execute LOS. The planner path is first conditioned under bounded-deviation and swept-footprint constraints, after which only local corner regions are modified. At each corner, the method derives at most two geometrically grounded trim distances d : a preferred value based on the adjacent edge lengths and a compatibility value that reserves at most half of the usable shared segment after a margin. For each d , the dimensionless ratio $\alpha=q/d$ is searched on a coarse grid from 0.1 to 0.5, refined between the neighbors of the winning sample, and recovered using a half-step grid only when all coarse samples fail. A quintic G^2 Bézier transition is admitted only if it satisfies geometric validity, non-reversing inner-wheel motion, kinematic limits, timing constraints, and true swept-footprint collision checks. A time gate and dynamic programming then select among transitions and in-place rotations while rejecting states whose trims overlap on a shared segment.

The evaluation contains 175 records from 7 Gazebo environments, 5 global planners, and 5 alternatives: Raw, Simple, Savitzky–Golay, Constrained, and PSTMO. Within each case, all alternatives receive the same Raw path. PSTMO returned a valid output in all 35/35 calls, executed exactly one `condition_only` pipeline without LOS, passed the final invariant checks, and produced no footprint-collision samples. Raw, Savitzky–Golay, and Constrained also succeeded in 35/35 cases, whereas Simple succeeded in 34/35. Across the 34 cases completed by all five alternatives, PSTMO achieved a mean path length of 10.1766 m, maximum curvature of 2.4047 m^{-1} , and curvature energy of 3.3121 m^{-1} . Relative to Simple, which attained the lowest mean ROS 2 baseline value for each corresponding metric, these quantities decreased by 0.64%, 75.29%, and 81.44%, respectively. Relative to Constrained, maximum curvature and curvature energy decreased by 80.62% and 92.10%. PSTMO generated 221 G^2 transitions and 6 pivots over 229 corners, with a mean minimum footprint clearance of 0.1708 m. The trade-off was a mean smoothing time of 95.21 ms, compared with 22.59 ms for Constrained. The results demonstrate the geometric effectiveness and footprint validity of corridor-preserving PSTMO; closed-loop goal-reaching performance remains to be quantified in execution and hardware experiments.

Keywords—PSTMO; differential-drive mobile robot; corridor-preserving path smoothing; G^2 Bézier transition; coarse-to-fine search; swept footprint; ROS 2/Nav2.